

SMAP CDF-matching sensitivity experiment over North Morocco

Domain: NorthMor | Year: 2016

Experiments Compared:

- OPL_noirr_2016
- DA_smap_nocdf_noirr_2016
- DA_smap_cdf_noirr_2016

Objective: Compare the impact of DA across experiments against the Open-Loop.

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1. Assimilation diagnostics

Figure 1.1 — Fig01 smap assimilated observation count

Assimilated SMAP observations in 2016

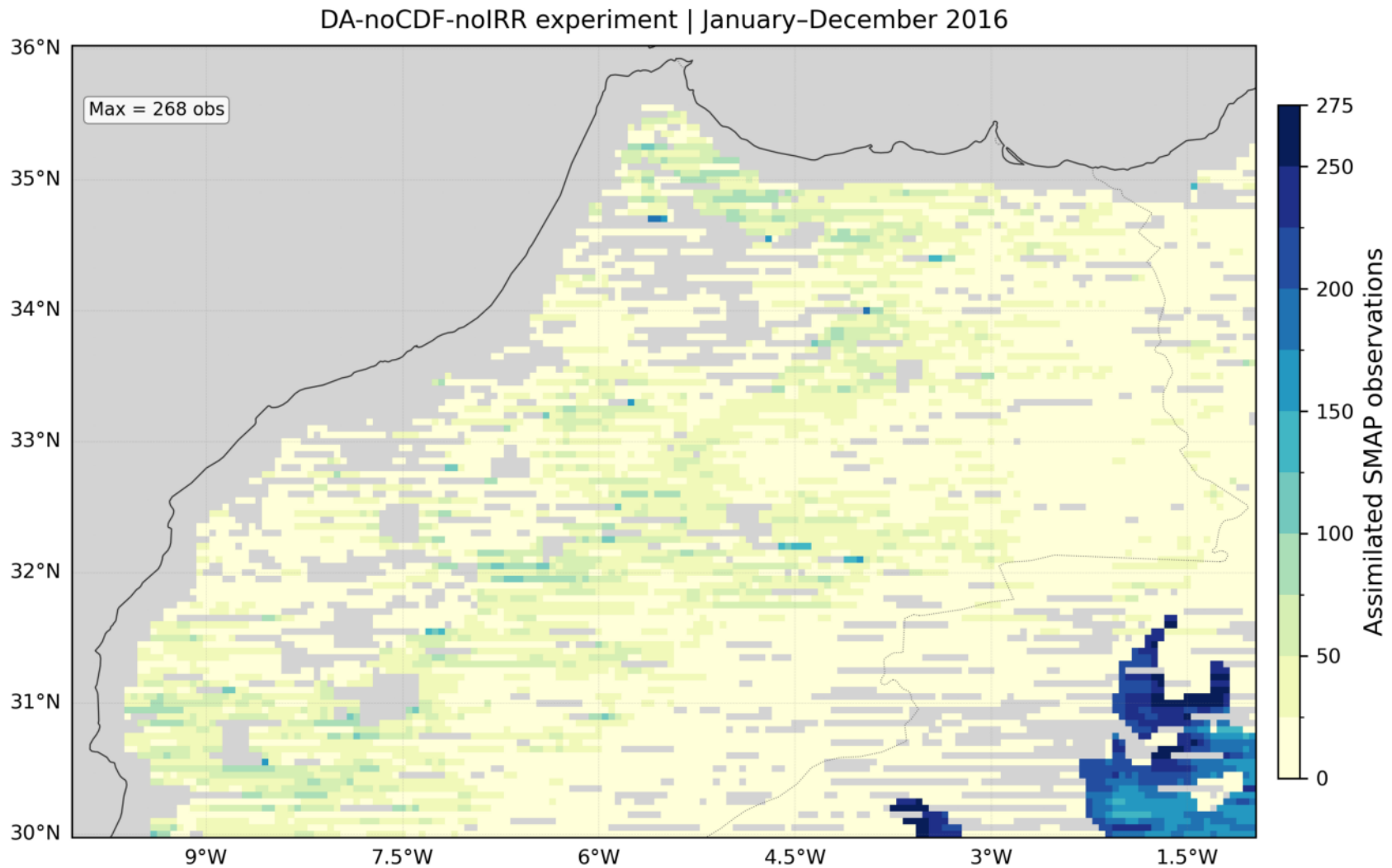
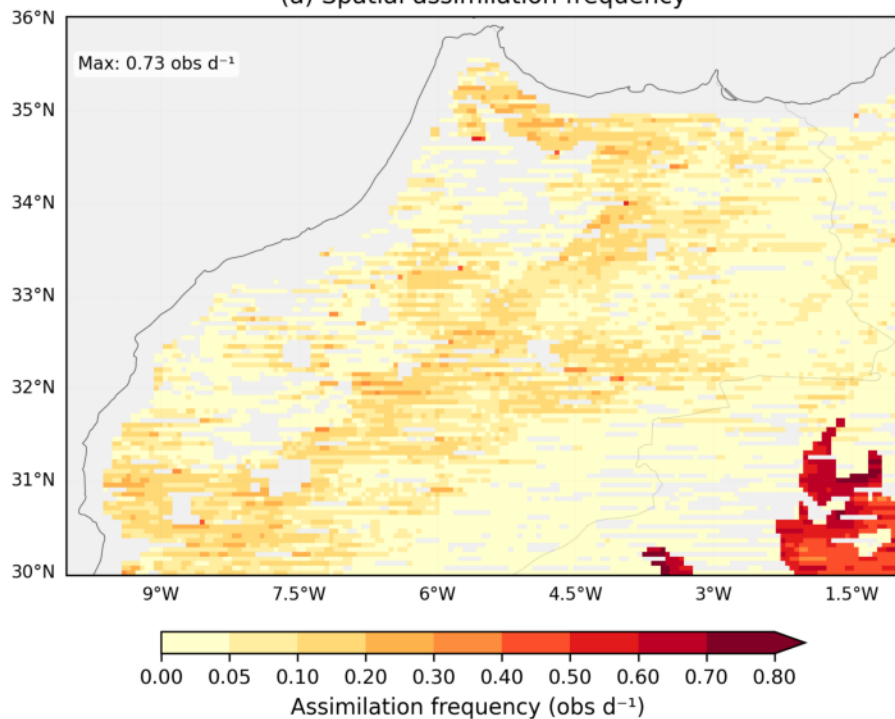


Figure 1.2 — Fig02 smap assimilation frequency

SMAP assimilation coverage in 2016

(a) Spatial assimilation frequency



(b) Monthly assimilated observations

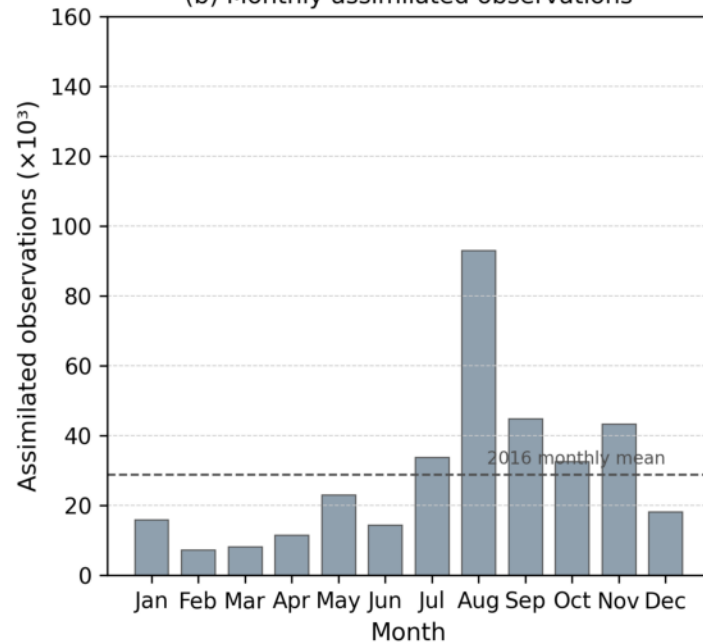


Figure 1.3 — Fig04a mean innovation observation space

Mean SMAP innovation in 2016

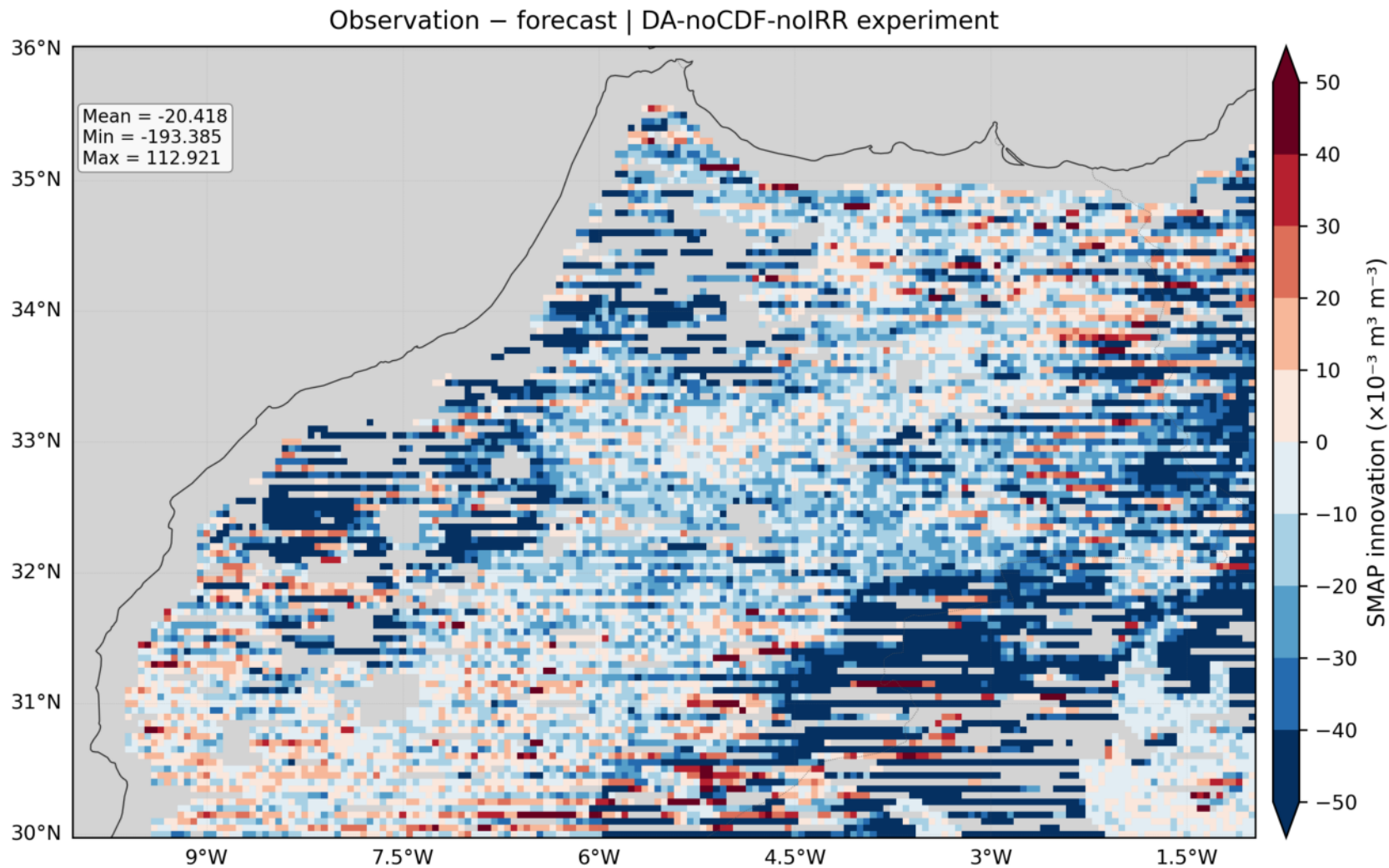


Figure 1.4 — Fig04b mean soil moisture increment

Mean SMAP analysis increment in 2016

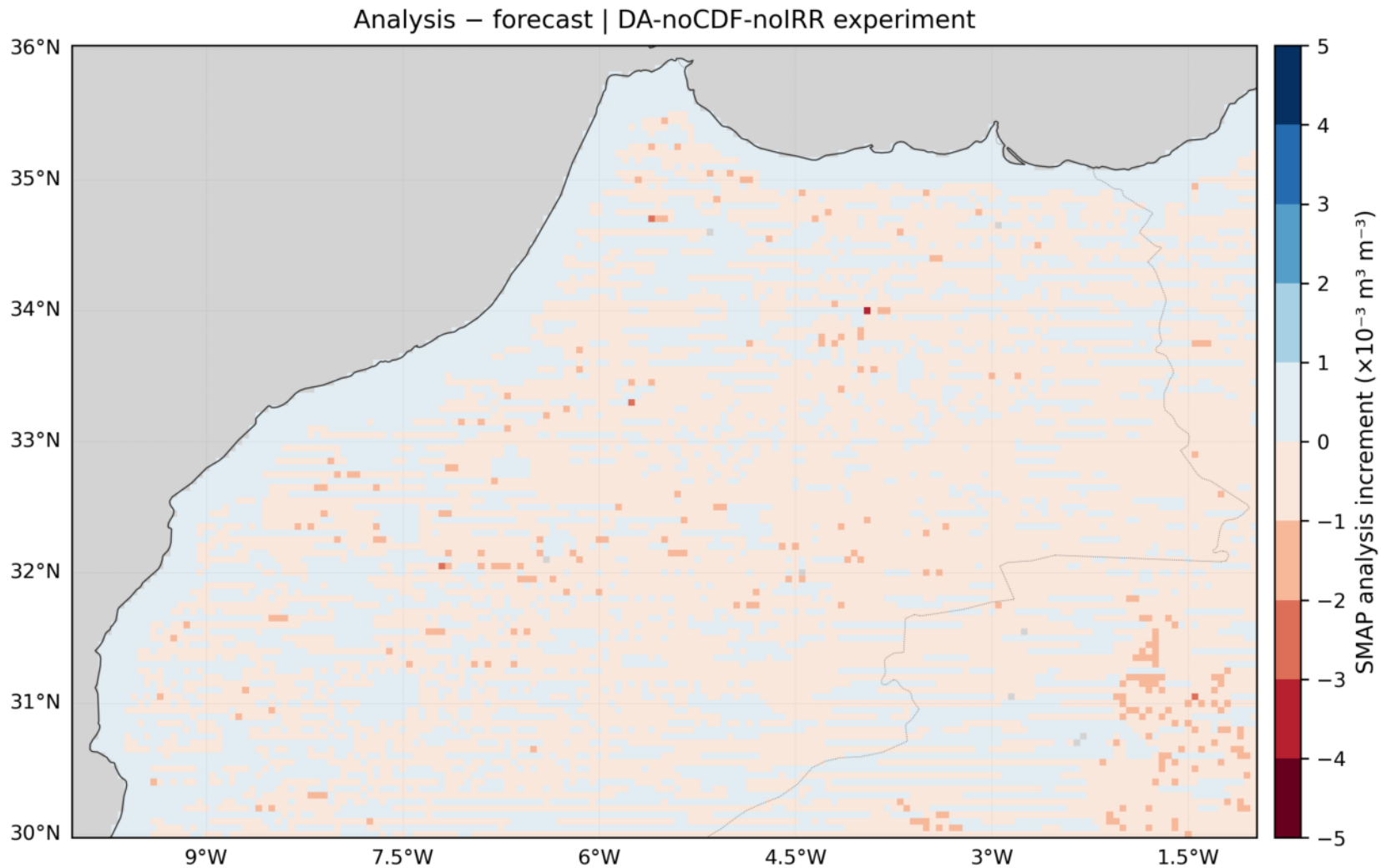


Figure 1.5 — Fig04c soil moisture increment distribution

Distribution of SMAP analysis increments in 2016

Analysis – forecast | DA-noCDF-noIRR experiment

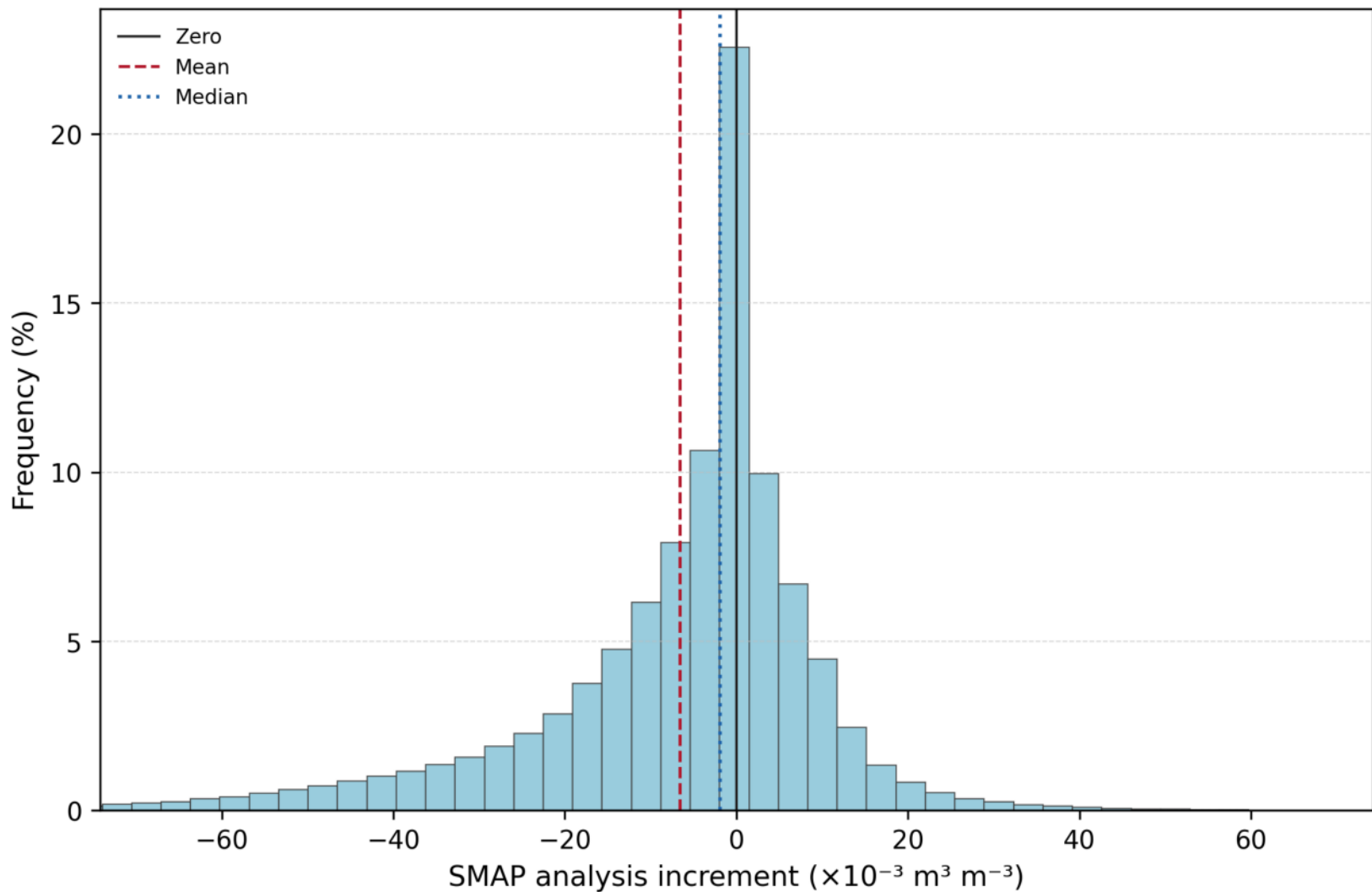


Figure 1.6 — Fig05 wet dry season increment comparison

Seasonal SMAP analysis increments in 2016

Analysis – forecast | DA-noCDF-noIRR experiment

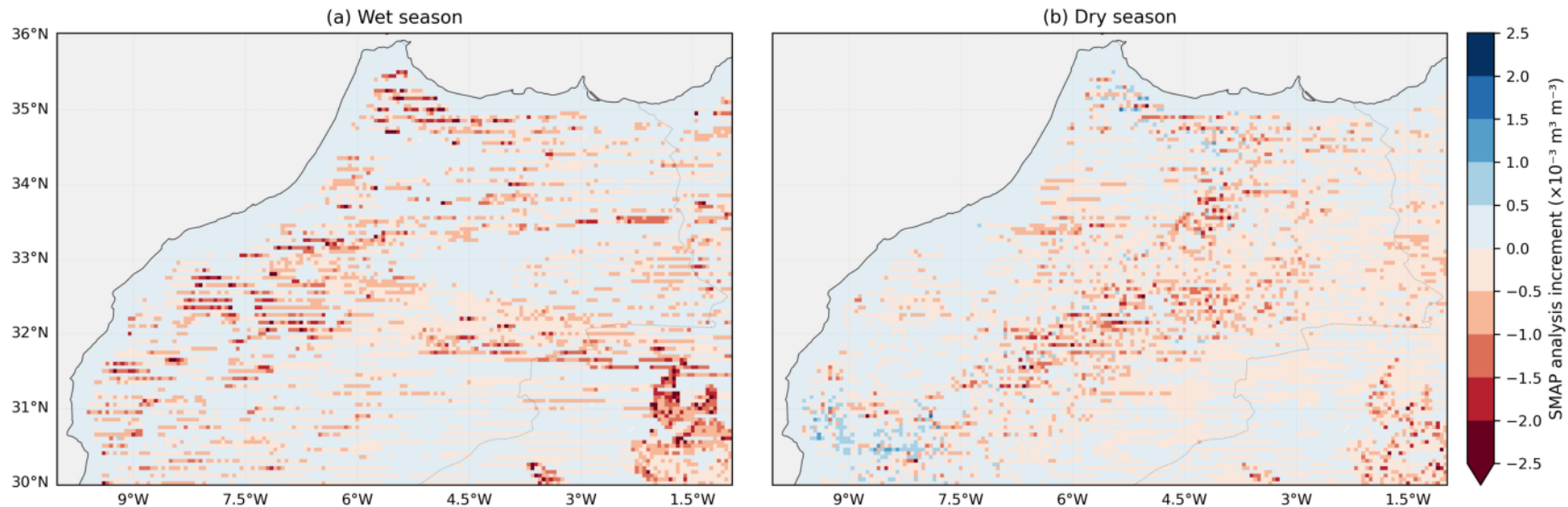


Figure 1.7 — Fig06 spread diagnostics consistency check

Spread diagnostic consistency check
forecast_sigma_01 and ensspread are not directly comparable | DA-noCDF-noIRR experiment

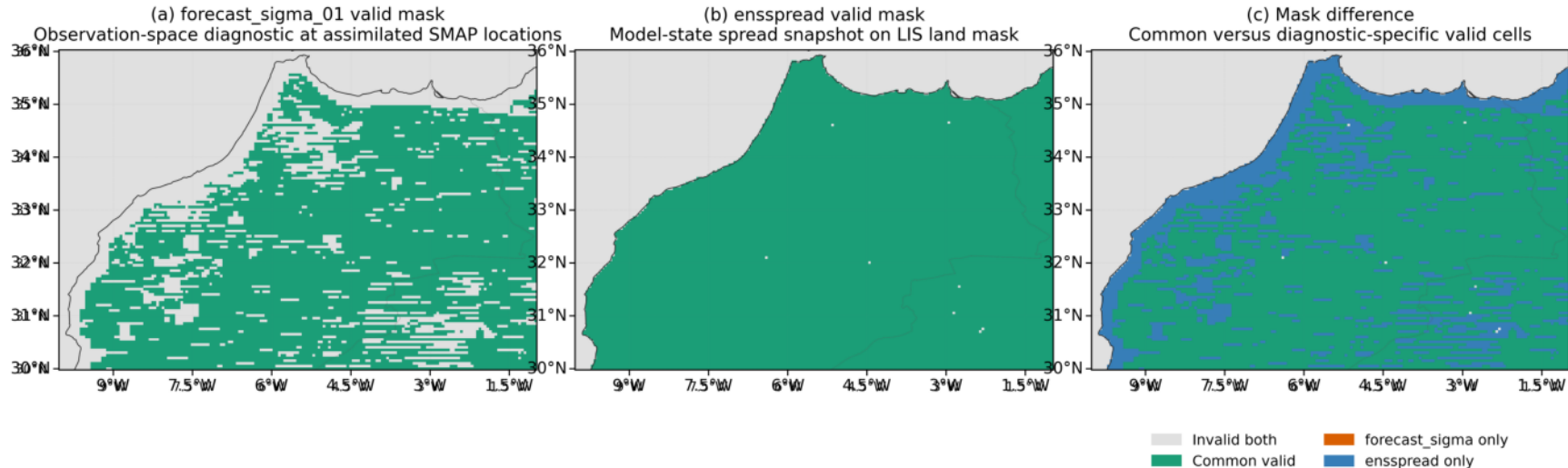


Figure 1.8 — Fig07 spread related diagnostics obsspace vs state

Spread-related diagnostics | DA-noCDF-noIRR, 2016

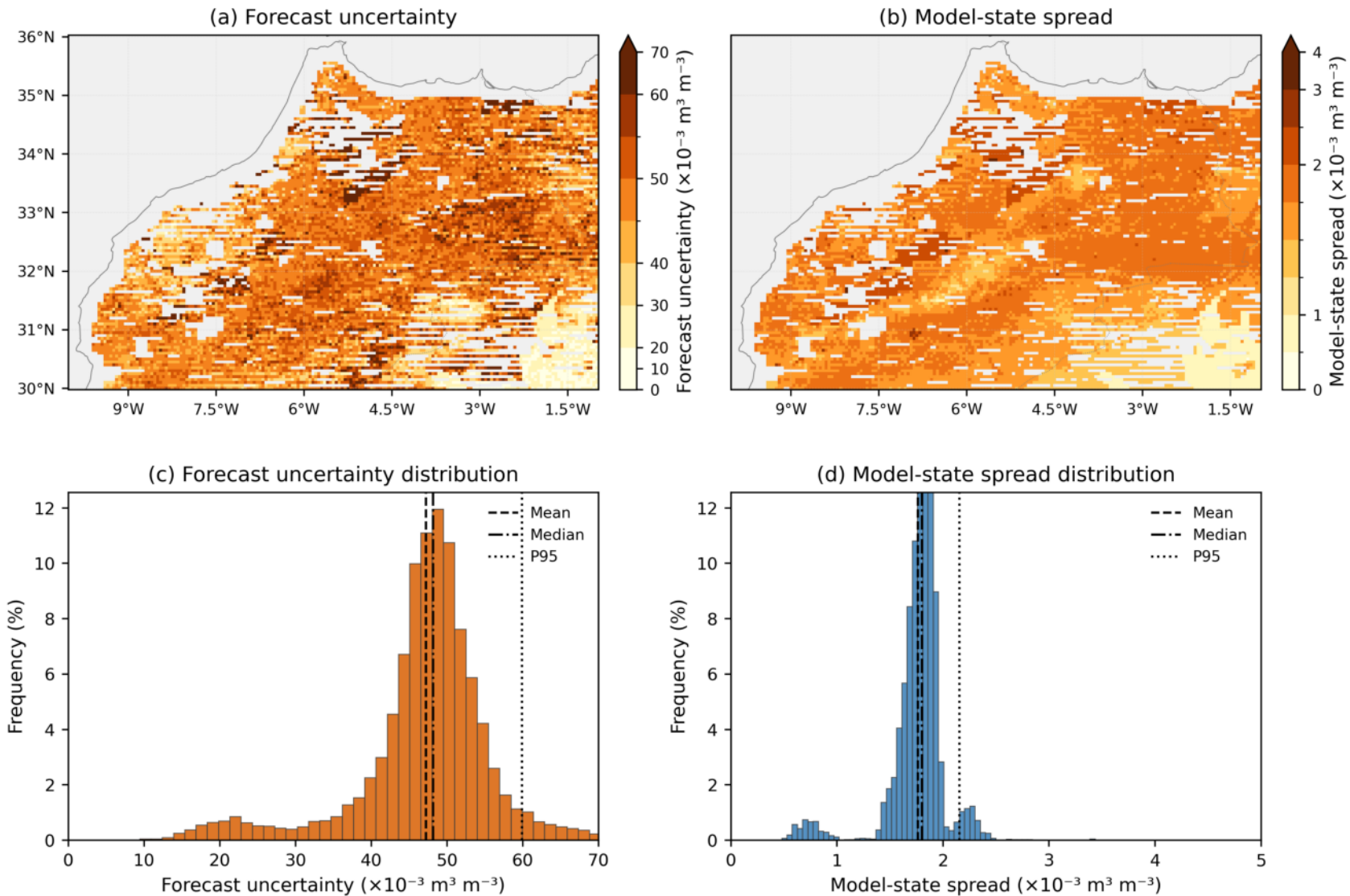


Figure 1.9 — Fig01 smap assimilated observation count

Assimilated SMAP observations in 2016

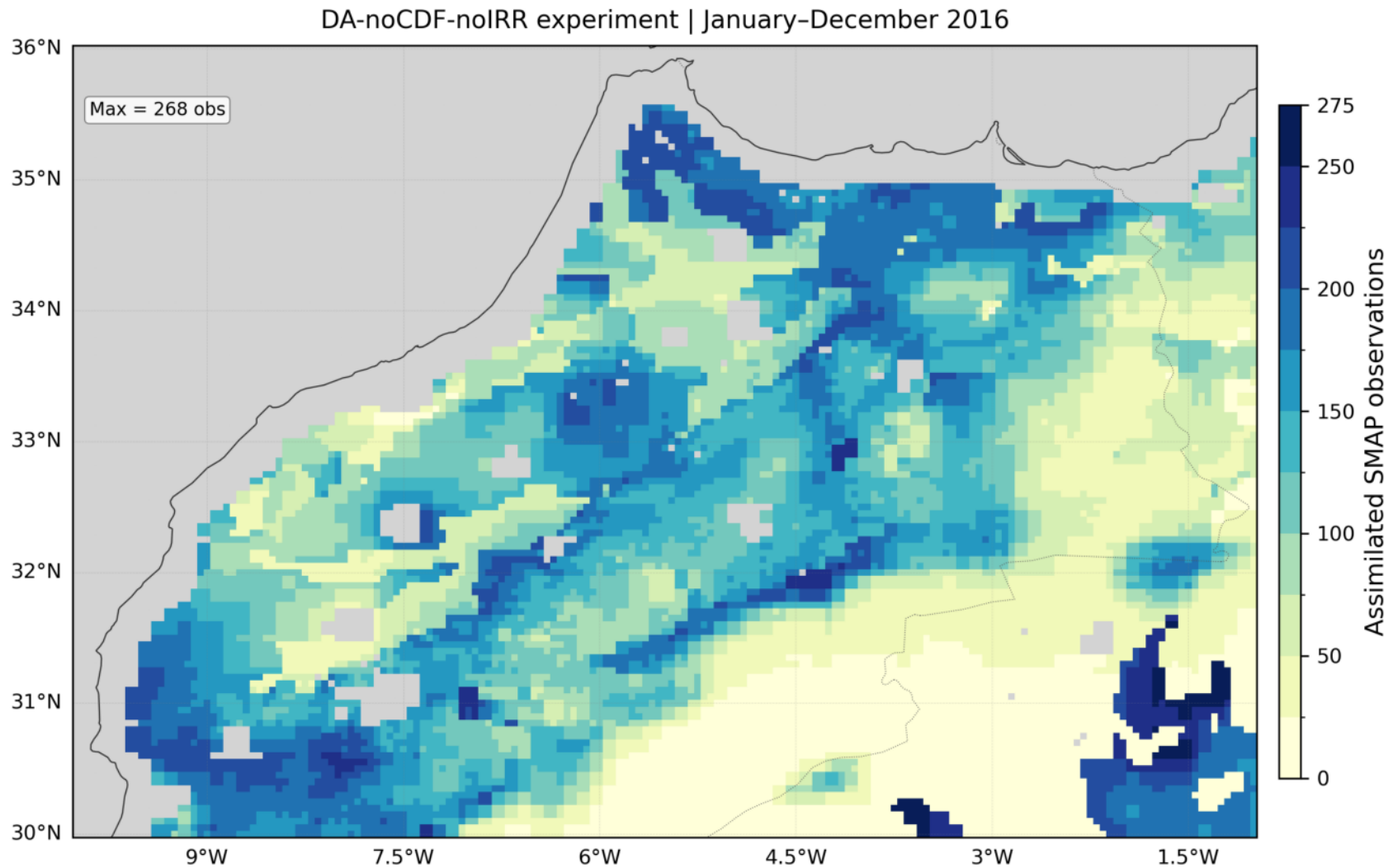
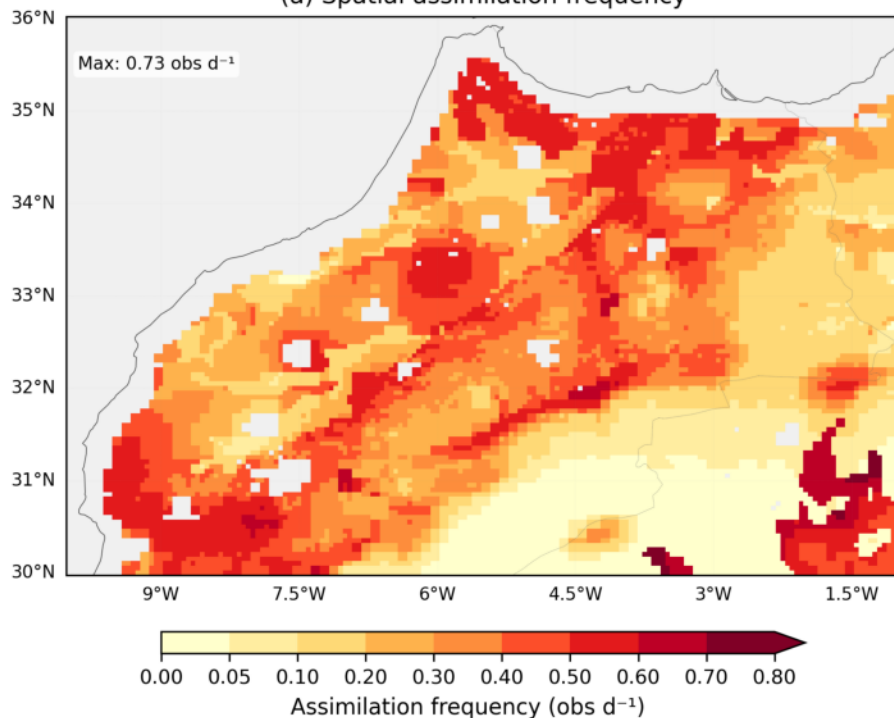


Figure 1.10 — Fig02 smap assimilation frequency

SMAP assimilation coverage in 2016

(a) Spatial assimilation frequency



(b) Monthly assimilated observations

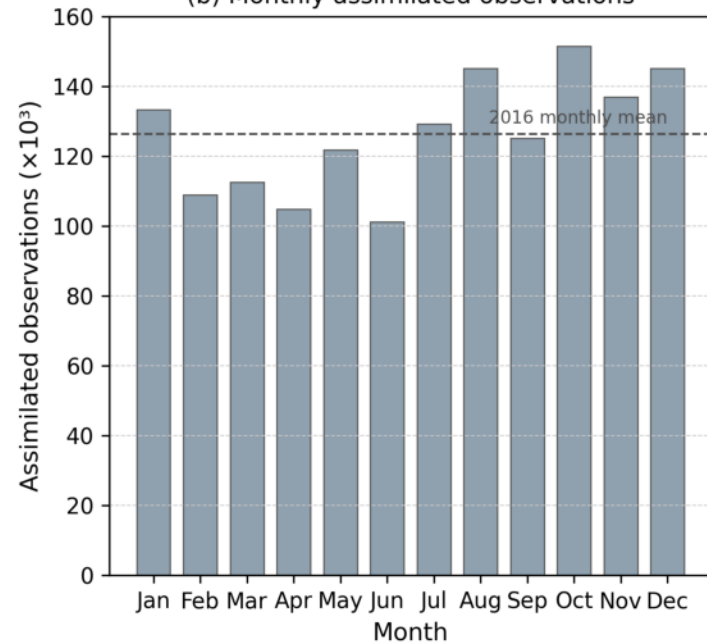


Figure 1.11 — Fig04a mean innovation observation space

Mean SMAP innovation in 2016

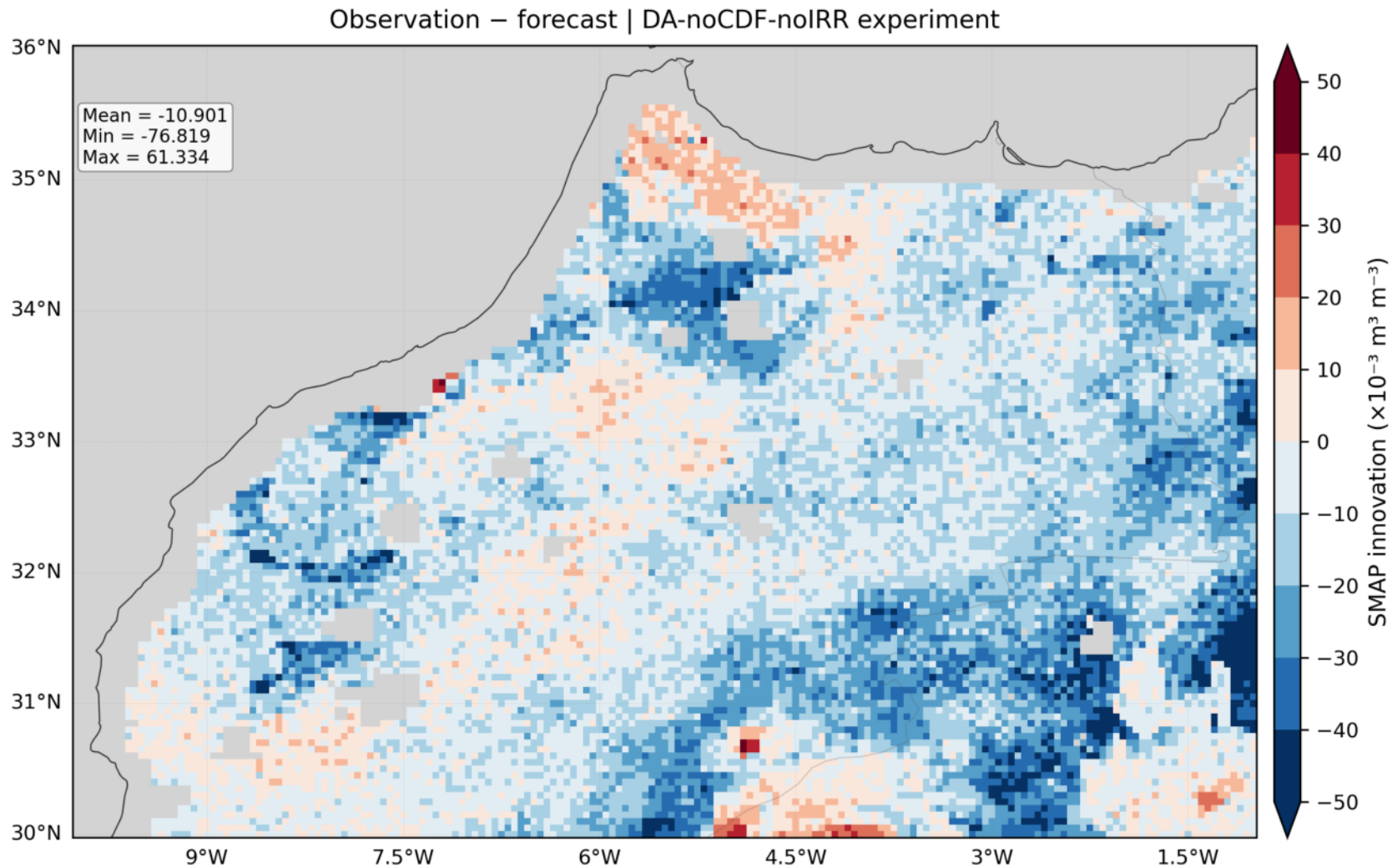


Figure 1.12 — Fig04b mean soil moisture increment

Mean SMAP analysis increment in 2016

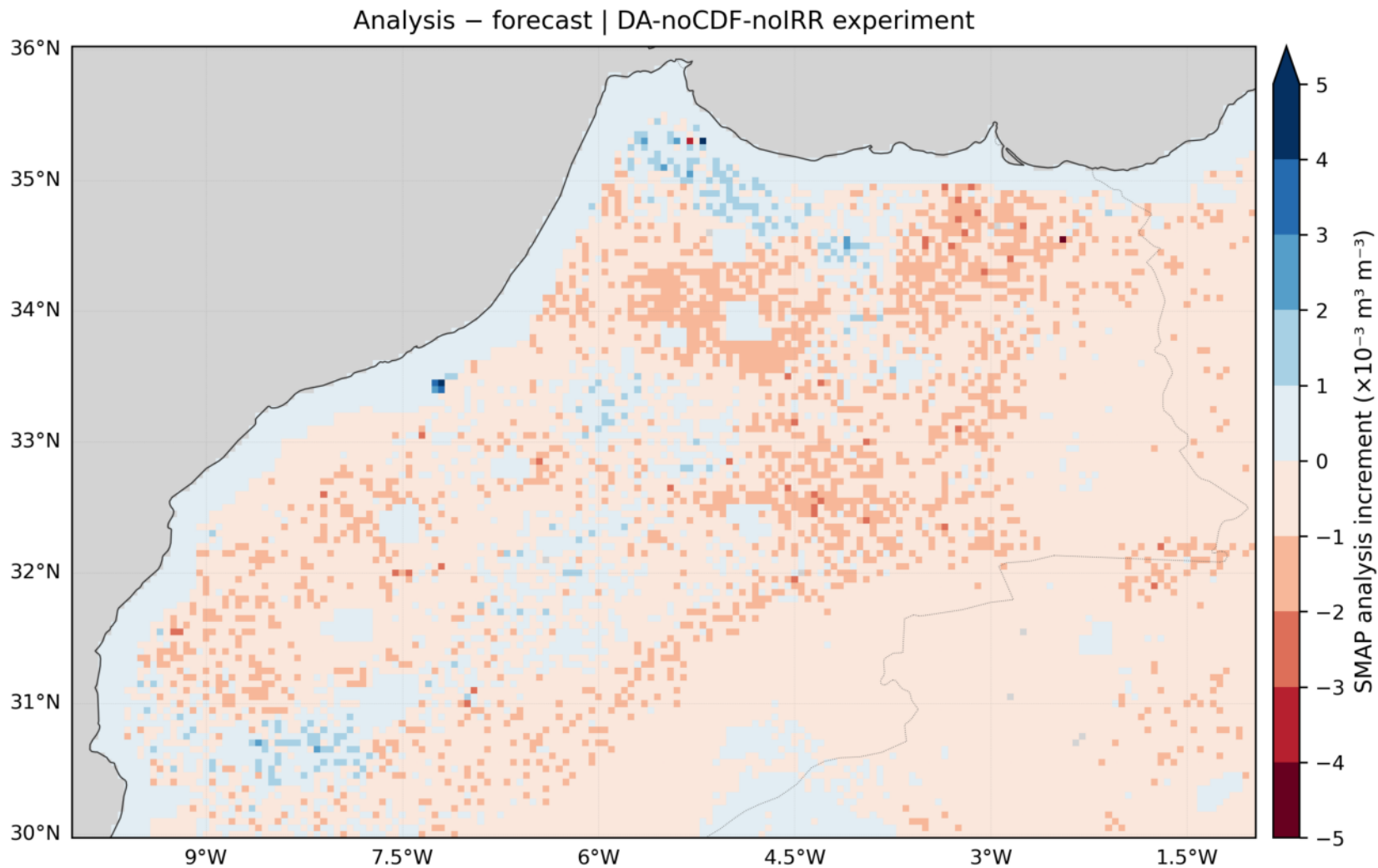


Figure 1.13 — Fig04c soil moisture increment distribution

Distribution of SMAP analysis increments in 2016

Analysis – forecast | DA-noCDF-noIRR experiment

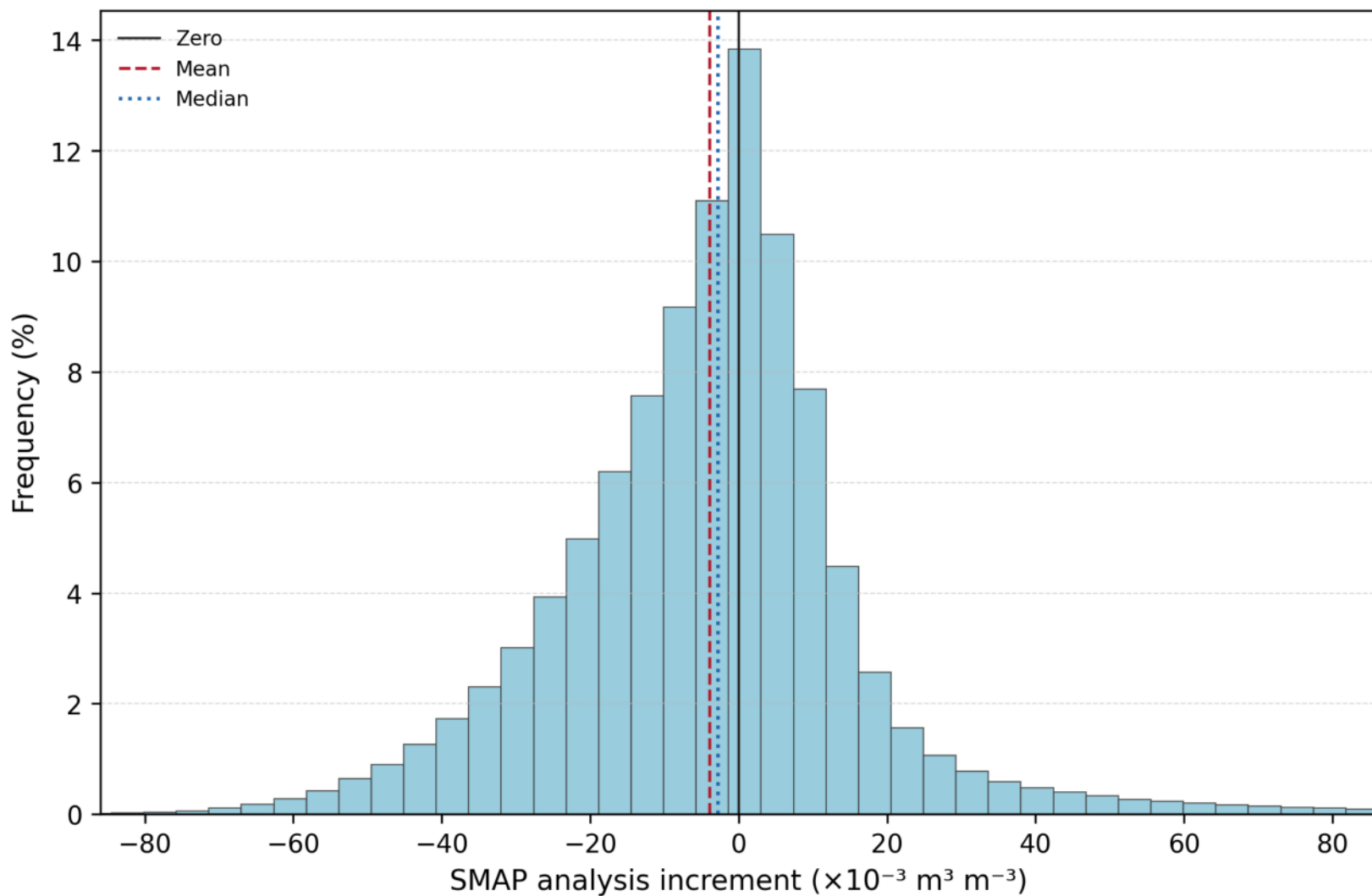


Figure 1.14 — Fig05 wet dry season increment comparison

Seasonal SMAP analysis increments in 2016

Analysis – forecast | DA-noCDF-noIRR experiment

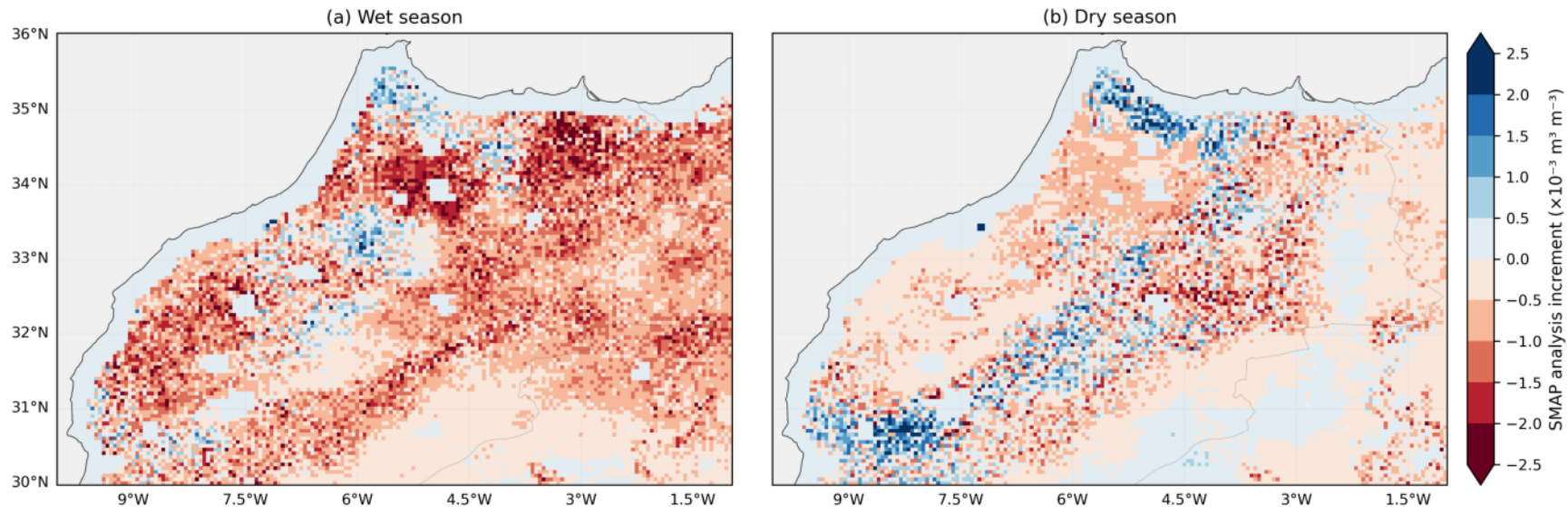


Figure 1.15 — Fig06 spread diagnostics consistency check

Spread diagnostic consistency check

forecast_sigma_01 and ensspread are not directly comparable | DA-noCDF-noIRR experiment

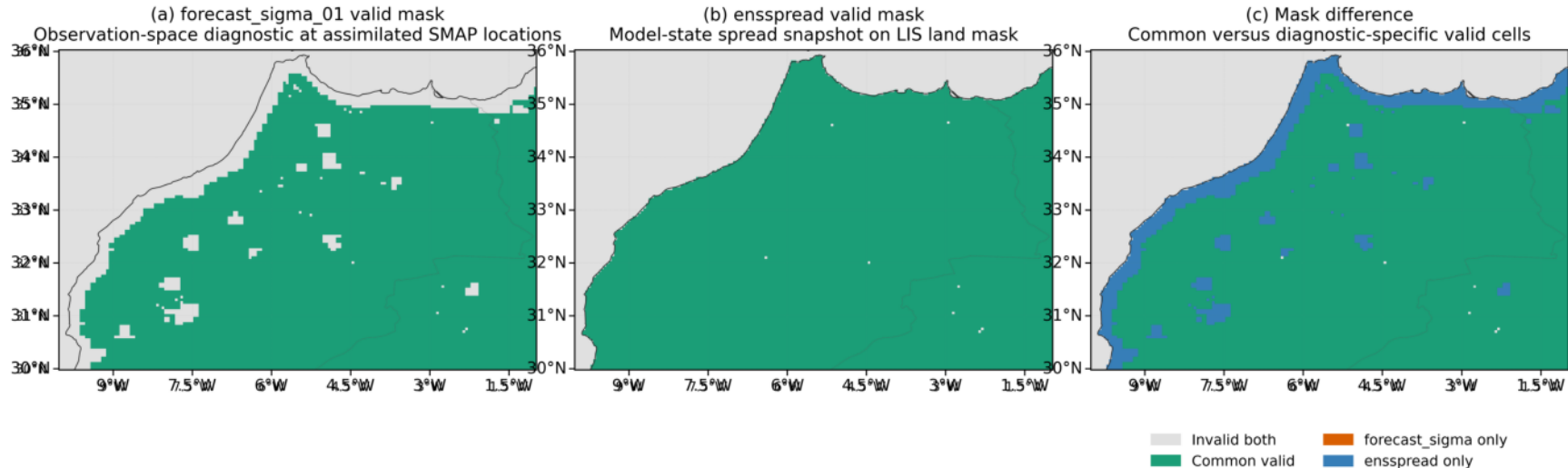
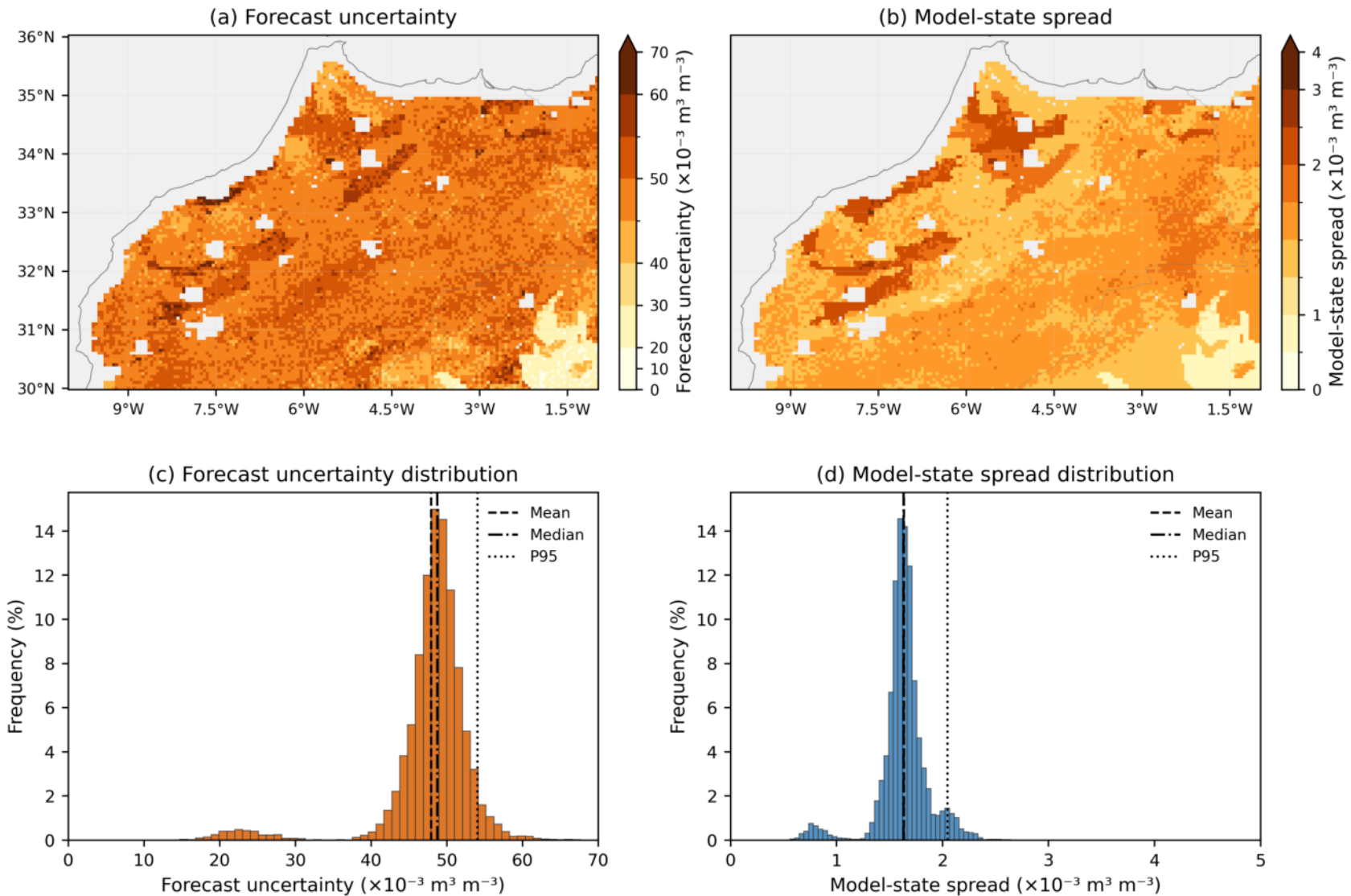


Figure 1.16 — Fig07 spread related diagnostics obsspace vs state

Spread-related diagnostics | DA-noCDF-noIRR, 2016



2. Surface soil moisture response

No figures available for Surface soil moisture response.

3. Root-zone and vertical soil moisture propagation

No figures available for Root-zone and vertical soil moisture propagation.

4. Surface fluxes

No figures available for Surface fluxes.

5. Groundwater response

No figures available for Groundwater response.

6. Runoff and baseflow response

No figures available for Runoff and baseflow response.

7. Water balance synthesis

Water balance synthesis figures not found in the output directory.

8. Spatial analysis

No figures available for Spatial analysis.

9. Streamflow validation

Streamflow validation figures were not generated for this run.
HyMAP routing and/or observed streamflow validation may be added in a later step.

10. External validation

External validation figures were not generated for this run.

Independent products such as WaPOR, ASCAT, ESA-CCI, GRACE or GLDAS may be added later.

Diagnostic summary checklist

- ☐ Are SMAP observations spatially well distributed?
- ☐ Is the CDF correction clearly modifying the assimilation signal?
- ☐ Are increments physically coherent?
- ☐ Does the DA signal propagate from surface to root-zone soil moisture?
- ☐ Is the ET response consistent with soil moisture changes?
- ☐ Is runoff response hydrologically plausible?
- ☐ Is baseflow response realistic?
- ☐ Are there regions with unrealistic DA impact?
- ☐ Which figures should be kept for the paper?
- ☐ Which figures should go to supplementary material?
- ☐ Which diagnostics need revision?